Step-by-step guide upon setting up SageMath and Jupyter Notebook/Lab

(Ensure internet connection for the whole process).

Part 1: Installation of Ubuntu

Ubuntu, on this guide, is necessary to successfully download and install SageMath. Here are the steps you should follow:

1. Press the Win key, search for Windows PowerShell, and open it.
2. On the PowerShell, type:

   >> wsl --install -d Ubuntu

3. (You might need to create an account first, so you'll need to create a new account by typing a new Username and Password. Note: Upon typing and re-typing your password, texts would not be visible.).
4. After the Ubuntu installation, close the PowerShell, search for Ubuntu and launch it.

Part 2: Downloading and Installing Conda (or Miniforge)

Conda is what we'll use to download, install, and create SageMath environments.

5. Type the following command on Ubuntu to download Miniforge (this is continuous text):

```
curl -L -O
"https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-$(uname
)-$(uname -m).sh"
```

You should see something like this:

```
(base) lester@LesterPC:~$ curl -L -O "https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-$(uname)-$(uname -m).sh"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100 81.7M  100 81.7M    0     0  19.8M      0  0:00:04  0:00:04 --:--:-- 28.3M
```

6. Install Miniforge using this command:

>> bash Miniforge3-$(uname)-$(uname -m).sh

You should see something like this:

```
(base) lester@LesterPC:~$ bash Miniforge3-$(uname)-$(uname -m).sh

Welcome to Miniforge3 25.11.0-1

In order to continue the installation process, please review the license
agreement.
Please, press ENTER to continue
>>>
Miniforge installer code uses BSD-3-Clause license as stated below.

Binary packages that come with it have their own licensing terms
and by installing miniforge you agree to the licensing terms of individual
packages as well. They include different OSI-approved licenses including
the GNU General Public License and can be found in pkgs/<pkg-name>/info/licenses
folders.

Miniforge installer comes with a bootstrapping executable that is used
when installing miniforge and is deleted after miniforge is installed.
The bootstrapping executable uses micromamba, cli11, cpp-filesystem,
curl, c-ares, krb5, libarchive, libev, lz4, nghttp2, openssl, libsolv,
nlohmann-json, reproc and zstd which are licensed under BSD-3-Clause,
MIT and OpenSSL licenses. Licenses and copyright notices of these
projects can be found at the following URL.
https://github.com/conda-forge/micromamba-feedstock/tree/master/recipe.

===============================================================================

Copyright (c) 2019-2022, conda-forge
All rights reserved.

Redistribution and use in source and binary forms, with or without
modification, are permitted provided that the following conditions are met:
```

7. Just click Enter until you reach:

```
1. Redistributions of source code must retain the above copyright notice, this
list of conditions and the following disclaimer.

2. Redistributions in binary form must reproduce the above copyright notice,
this list of conditions and the following disclaimer in the documentation
and/or other materials provided with the distribution.

3. Neither the name of the copyright holder nor the names of its contributors
may be used to endorse or promote products derived from this software without
specific prior written permission.

THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND
ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED
WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE
DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT HOLDER OR CONTRIBUTORS BE LIABLE
FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL
DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR
SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER
CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY,
OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.


Do you accept the license terms? [yes|no]
>>> yes
```

8. Type and enter: yes

9. Always type yes (or y) if asked to.

10. You'll know that you have successfully downloaded and installed Miniforge if you see this

```
Proceed with initialization? [yes|no]
[no] >>> yes
no change     /home/lester/miniforge3/condabin/conda
no change     /home/lester/miniforge3/bin/conda
no change     /home/lester/miniforge3/bin/conda-env
no change     /home/lester/miniforge3/bin/activate
no change     /home/lester/miniforge3/bin/deactivate
no change     /home/lester/miniforge3/etc/profile.d/conda.sh
no change     /home/lester/miniforge3/etc/fish/conf.d/conda.fish
no change     /home/lester/miniforge3/shell/condabin/Conda.psm1
no change     /home/lester/miniforge3/shell/condabin/conda-hook.ps1
no change     /home/lester/miniforge3/lib/python3.12/site-packages/xontrib/conda.xsh
no change     /home/lester/miniforge3/etc/profile.d/conda.csh
modified      /home/lester/.bashrc

==> For changes to take effect, close and re-open your current shell. <==

Running `shell init`, which:
 - modifies RC file: "/home/lester/.bashrc"
 - generates config for root prefix: "/home/lester/miniforge3"
 - sets mamba executable to: "/home/lester/miniforge3/bin/mamba"
The following has been added in your "/home/lester/.bashrc" file

# >>> mamba initialize >>>
# !! Contents within this block are managed by 'mamba shell init' !!
export MAMBA_EXE='/home/lester/miniforge3/bin/mamba';
export MAMBA_ROOT_PREFIX='/home/lester/miniforge3';
__mamba_setup="$("$MAMBA_EXE" shell hook --shell bash --root-prefix "$MAMBA_ROOT_PREFIX" 2> /dev/null)"
if [ $? -eq 0 ]; then
    eval "$__mamba_setup"
else
    alias mamba="$MAMBA_EXE"  # Fallback on help from mamba activate
fi
unset __mamba_setup
# <<< mamba initialize <<<

Thank you for installing Miniforge3!
(base) lester@LesterPC:~$
```

11. Close and re-open Ubuntu.

Part 3: Creating Sage Environment:

In this part, we'll create a Sage environment inside Conda.

12. Upon opening Ubuntu, you should see (base), meaning your Conda is

    activated. If you didn't see one, just type (otherwise, skip this step):

    >> conda activate

13. Create Sage environment using Python 3.12 by typing:

    >> conda create -n sage python=3.12

```
(base) lester@LesterPC:~$ conda create -n sage python=3.12
Retrieving notices: done
Channels:
 - conda-forge
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done


==> WARNING: A newer version of conda exists. <==
    current version: 25.11.0
    latest version: 26.1.0

Please update conda by running

    $ conda update -n base -c conda-forge conda



## Package Plan ##

  environment location: /home/lester/miniforge3/envs/sage

  added / updated specs:
    - python=3.12


The following packages will be downloaded:

    package                    |            build
    ---------------------------|-----------------
    ca-certificates-2026.1.4   |       hbd8a1cb_0         143 KB  conda-forge
    icu-78.2                   |       h33c6efd_0        12.1 MB  conda-forge
    ld_impl_linux-64-2.45.1    |default_hbd61a6d_101         709 KB  conda-forge
    libffi-3.5.2               |       h3435931_0          57 KB  conda-forge
    libgcc-15.2.0              |      he0feb66_17        1016 KB  conda-forge
    libgcc-ng-15.2.0           |      h69a702a_17          27 KB  conda-forge
    libgomp-15.2.0             |      he0feb66_17         589 KB  conda-forge
    liblzma-5.8.2              |       hb03c661_0         111 KB  conda-forge
    libsqlite-3.51.2           |       hf4e2dac_0         921 KB  conda-forge
    libstdcxx-15.2.0           |      h934c35e_17         5.6 MB  conda-forge
```

14. Always press yes if you are asked.

15. After the process, activate the created Sage environment by typing:

    >> conda activate sage

Part 4: Installation of Sage

16. Type this command to install Sage:

    >> conda install sage -c conda-forge

17. Press y (or yes) if asked. Wait until everything finishes.

18. You'll know that sage is installed if you its version upon entering:

    >> sage --version

You must see this:

```
Proceed ([y]/n)? y


Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
(sage) lester@LesterPC:~$ sage --version
SageMath version 10.7, Release Date: 2025-08-09
(sage) lester@LesterPC:~$ |
```

Part 5: Register SageMath as a kernel in ipykernel for Jupyter to read

After installing SageMath, jumping right through Jupyter Notebook/Lab (if you have already installed it) will not display SageMath, so you can't use its feature. We must link SageMath to ipykernel first, it is the kernel that Jupyter Notebook/Lab reads to have the ability to execute things.

19. First, go out from the (sage) environment by typing >> exit
20. Type this command to register Sage as a kernel:

    >> sage -python -m ipykernel install --user --name=sage --display-name "SageMath"

Part 6: Installation of Jupyter Notebook/Lab

21. On Ubuntu, if you see something like (base), then we need to deactivate Conda. Deactivate Conda using:

    >> conda deactivate

22. You'll now see it no more. Check if python3 and pip3 is available in you PC by typing the commands:

    >> python3 --version

    >> pip3 --version

You must see something like this:

```
(sage) lester@LesterPC:~$ python3 --version
Python 3.12.12
(sage) lester@LesterPC:~$ pip3 --version
pip 26.0.1 from /home/lester/miniforge3/envs/sage/lib/python3.12/site-packages/pip (python 3.12)
```

Otherwise, you must install them by typing:

>> sudo apt update

>> sudo apt install python3 python3-pip

23. Go back to Step **22** for checking, then you can proceed and the installment is confirmed.

24. To install Jupyter Notebook, type:

>> pip3 install notebook

You must see this:

```
(sage) lester@LesterPC:~$ pip3 install notebook
Collecting notebook
  Downloading notebook-7.5.3-py3-none-any.whl.metadata (10 kB)
Collecting jupyter-server<3,>=2.4.0 (from notebook)
  Using cached jupyter_server-2.17.0-py3-none-any.whl.metadata (8.5 kB)
Collecting jupyterlab-server<3,>=2.28.0 (from notebook)
  Using cached jupyterlab_server-2.28.0-py3-none-any.whl.metadata (5.9 kB)
Collecting jupyterlab<4.6,>=4.5.3 (from notebook)
  Using cached jupyterlab-4.5.3-py3-none-any.whl.metadata (16 kB)
Collecting notebook-shim<0.3,>=0.2 (from notebook)
  Using cached notebook_shim-0.2.4-py3-none-any.whl.metadata (4.0 kB)
Collecting tornado>=6.2.0 (from notebook)
  Using cached tornado-6.5.4-cp39-abi3-manylinux_2_5_x86_64.manylinux1_x86_64.manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (2.8 kB)
Collecting anyio>=3.1.0 (from jupyter-server<3,>=2.4.0->notebook)
  Using cached anyio-4.12.1-py3-none-any.whl.metadata (4.3 kB)
Collecting argon2-cffi>=21.1 (from jupyter-server<3,>=2.4.0->notebook)
  Using cached argon2_cffi-25.1.0-py3-none-any.whl.metadata (4.1 kB)
Collecting jinja2>=3.0.3 (from jupyter-server<3,>=2.4.0->notebook)
  Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting jupyter-client>=7.4.4 (from jupyter-server<3,>=2.4.0->notebook)
  Using cached jupyter_client-8.8.0-py3-none-any.whl.metadata (8.4 kB)
Collecting jupyter-core!=5.0.*,>=4.12 (from jupyter-server<3,>=2.4.0->notebook)
  Using cached jupyter_core-5.9.1-py3-none-any.whl.metadata (1.5 kB)
Collecting jupyter-events>=0.11.0 (from jupyter-server<3,>=2.4.0->notebook)
  Using cached jupyter_events-0.12.0-py3-none-any.whl.metadata (5.8 kB)
Collecting jupyter-server-terminals>=0.4.4 (from jupyter-server<3,>=2.4.0->notebook)
  Using cached jupyter_server_terminals-0.5.4-py3-none-any.whl.metadata (5.9 kB)
Collecting nbconvert>=6.4.4 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading nbconvert-7.17.0-py3-none-any.whl.metadata (8.4 kB)
Collecting nbformat>=5.3.0 (from jupyter-server<3,>=2.4.0->notebook)
  Using cached nbformat-5.10.4-py3-none-any.whl.metadata (3.6 kB)
Requirement already satisfied: packaging>=22.0 in ./miniforge3/envs/sage/lib/python3.12/site-packages (from jupyter-server<3,>=2.4.0->notebook) (26.0)
Collecting prometheus-client>=0.9 (from jupyter-server<3,>=2.4.0->notebook)
  Using cached prometheus_client-0.24.1-py3-none-any.whl.metadata (2.1 kB)
Collecting pyzmq>=24 (from jupyter-server<3,>=2.4.0->notebook)
  Using cached pyzmq-27.1.0-cp312-abi3-manylinux_2_26_x86_64.manylinux_2_28_x86_64.whl.metadata (6.0 kB)
Collecting send2trash>=1.8.2 (from jupyter-server<3,>=2.4.0->notebook)
  Using cached send2trash-2.1.0-py3-none-any.whl.metadata (4.1 kB)
Collecting terminado>=0.8.3 (from jupyter-server<3,>=2.4.0->notebook)
  Using cached terminado-0.18.1-py3-none-any.whl.metadata (5.8 kB)
```

25. …or Jupyter Lab:

>> pip3 install jupyterlab

26. Close and re-open Ubuntu again, afterwards.

Part 7: Start Jupyter Notebook and use SageMath

27. Activate Sage by typing:

>> conda activate sage

28. Start Jupyter Notebook by typing:

>> jupyter notebook

29. …or Jupyter Lab by typing:

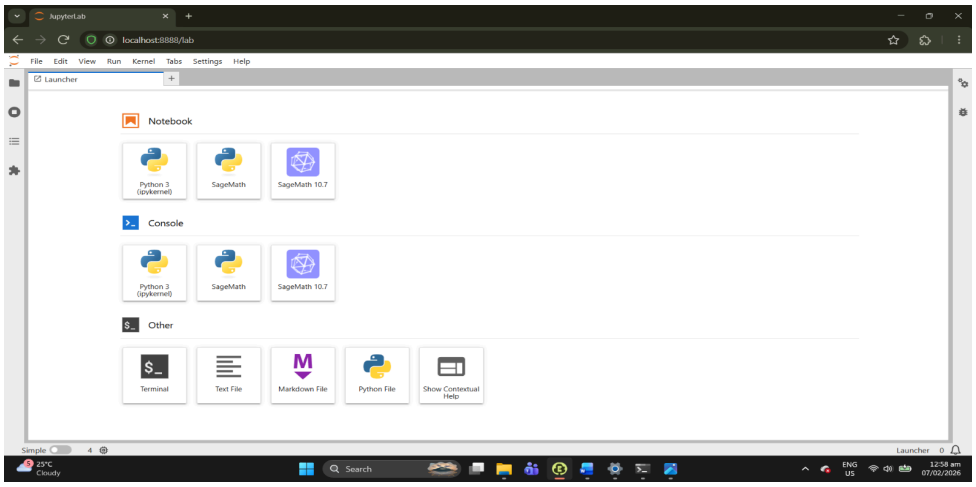
>> jupyter lab

You must see this:

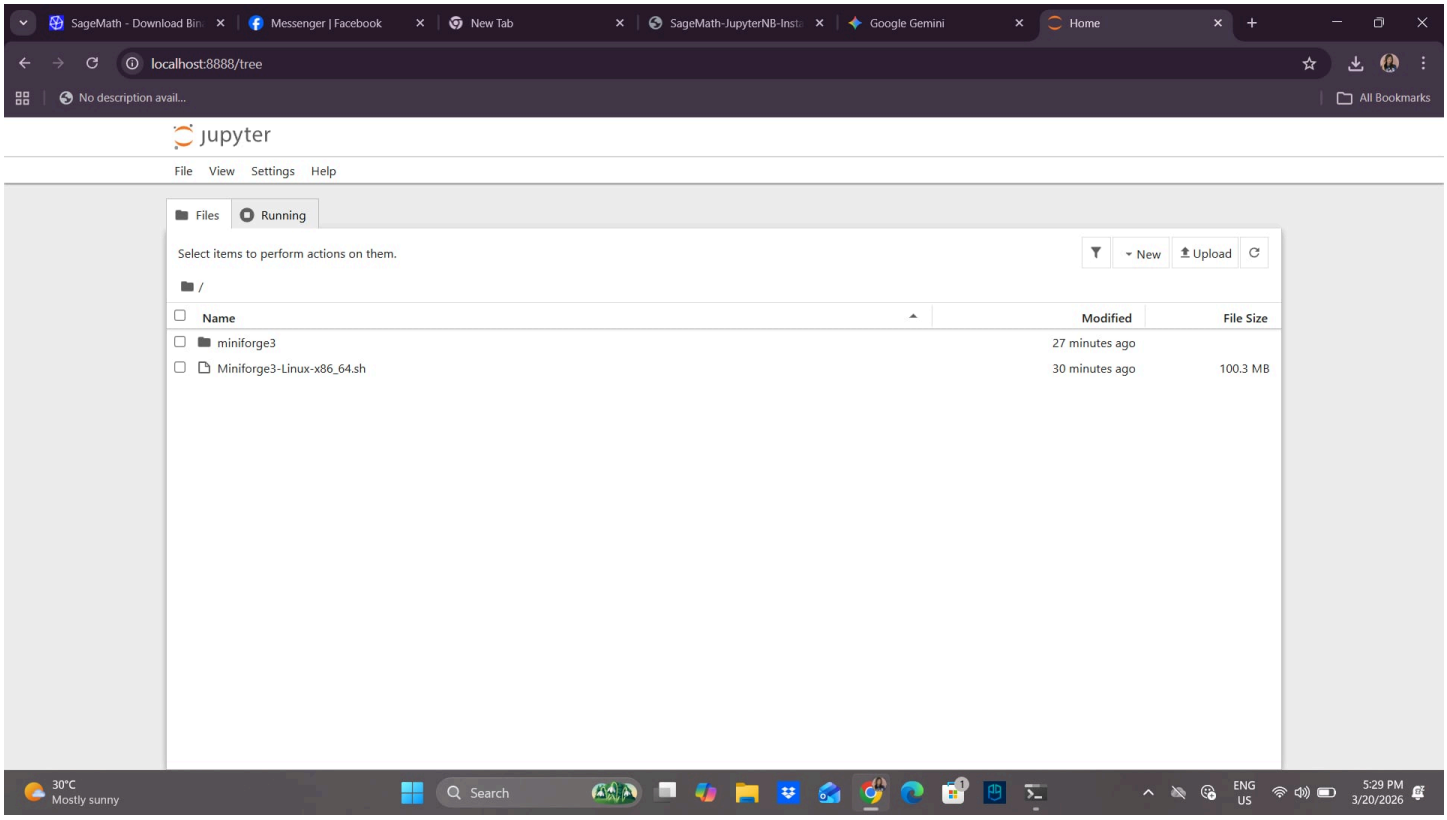
```
(base) lester@LesterPC:~$ conda activate sage
(sage) lester@LesterPC:~$ jupyter lab
[I 2026-02-06 23:33:33.801 ServerApp] jupyter_lsp | extension was successfully linked.
[I 2026-02-06 23:33:33.806 ServerApp] jupyter_server_terminals | extension was successfully linked.
[I 2026-02-06 23:33:33.812 ServerApp] jupyterlab | extension was successfully linked.
[I 2026-02-06 23:33:33.817 ServerApp] notebook | extension was successfully linked.
[I 2026-02-06 23:33:34.171 ServerApp] notebook_shim | extension was successfully linked.
[I 2026-02-06 23:33:34.191 ServerApp] notebook_shim | extension was successfully loaded.
[I 2026-02-06 23:33:34.194 ServerApp] jupyter_lsp | extension was successfully loaded.
[I 2026-02-06 23:33:34.195 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[I 2026-02-06 23:33:34.201 LabApp] JupyterLab extension loaded from /home/lester/miniforge3/envs/sage/lib/python3.12/site-packages/jupyterlab
[I 2026-02-06 23:33:34.201 LabApp] JupyterLab application directory is /home/lester/miniforge3/envs/sage/share/jupyter/lab
[I 2026-02-06 23:33:34.201 LabApp] Extension Manager is 'pypi'.
[I 2026-02-06 23:33:34.265 ServerApp] jupyterlab | extension was successfully loaded.
[I 2026-02-06 23:33:34.271 ServerApp] notebook | extension was successfully loaded.
[I 2026-02-06 23:33:34.273 ServerApp] Serving notebooks from local directory: /home/lester
[I 2026-02-06 23:33:34.273 ServerApp] Jupyter Server 2.17.0 is running at:
[I 2026-02-06 23:33:34.273 ServerApp] http://localhost:8888/lab?token=9b7979beb5c05e00c01e1edebddc2c0312352789b0a80d8b
[I 2026-02-06 23:33:34.273 ServerApp]     http://127.0.0.1:8888/lab?token=9b7979beb5c05e00c01e1edebddc2c0312352789b0a80d8b
[I 2026-02-06 23:33:34.273 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 2026-02-06 23:33:36.859 ServerApp]

    To access the server, open this file in a browser:
        file:///home/lester/.local/share/jupyter/runtime/jpserver-1483-open.html
    Or copy and paste one of these URLs:
        http://localhost:8888/lab?token=9b7979beb5c05e00c01e1edebddc2c0312352789b0a80d8b
        http://127.0.0.1:8888/lab?token=9b7979beb5c05e00c01e1edebddc2c0312352789b0a80d8b
WARNING: You don't seem to have any mimeinfo.cache files.
Try running the update-desktop-database command. If you
don't have this command you should install the
desktop-file-utils package. This package is available from
http://freedesktop.org/wiki/Software/desktop-file-utils/
No applications found for mimetype: text/html
./usr/bin/xdg-open: 882: x-www-browser: not found
/usr/bin/xdg-open: 882: firefox: not found
/usr/bin/xdg-open: 882: iceweasel: not found
/usr/bin/xdg-open: 882: seamonkey: not found
/usr/bin/xdg-open: 882: mozilla: not found
/usr/bin/xdg-open: 882: epiphany: not found
/usr/bin/xdg-open: 882: konqueror: not found
/usr/bin/xdg-open: 882: chromium: not found
```

30. Copy a link from any of the two links (highlighted by white), then paste it on any of your available browsers

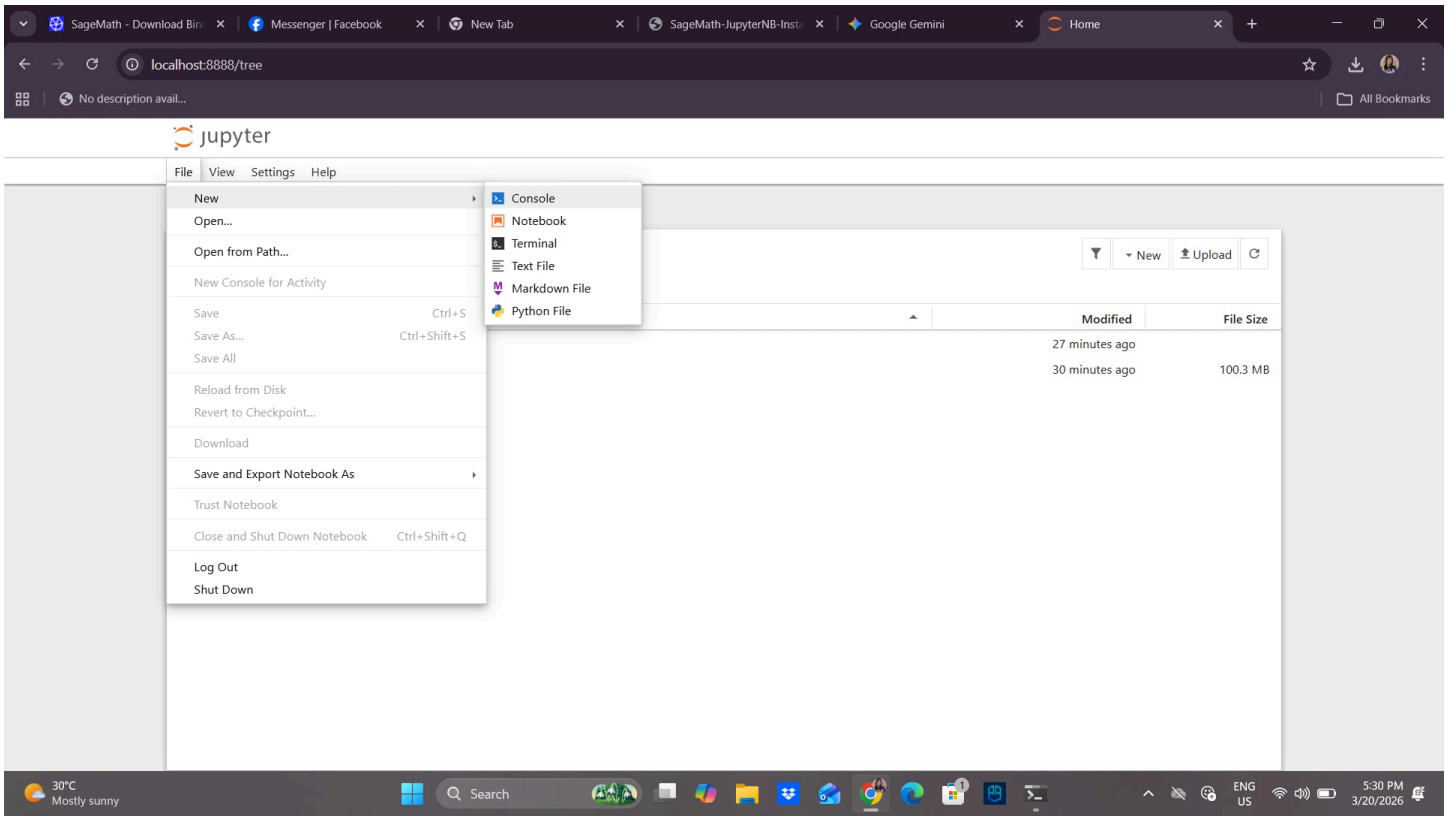
31. Then they live happily ever after. (The image below is Jupyter Lab)



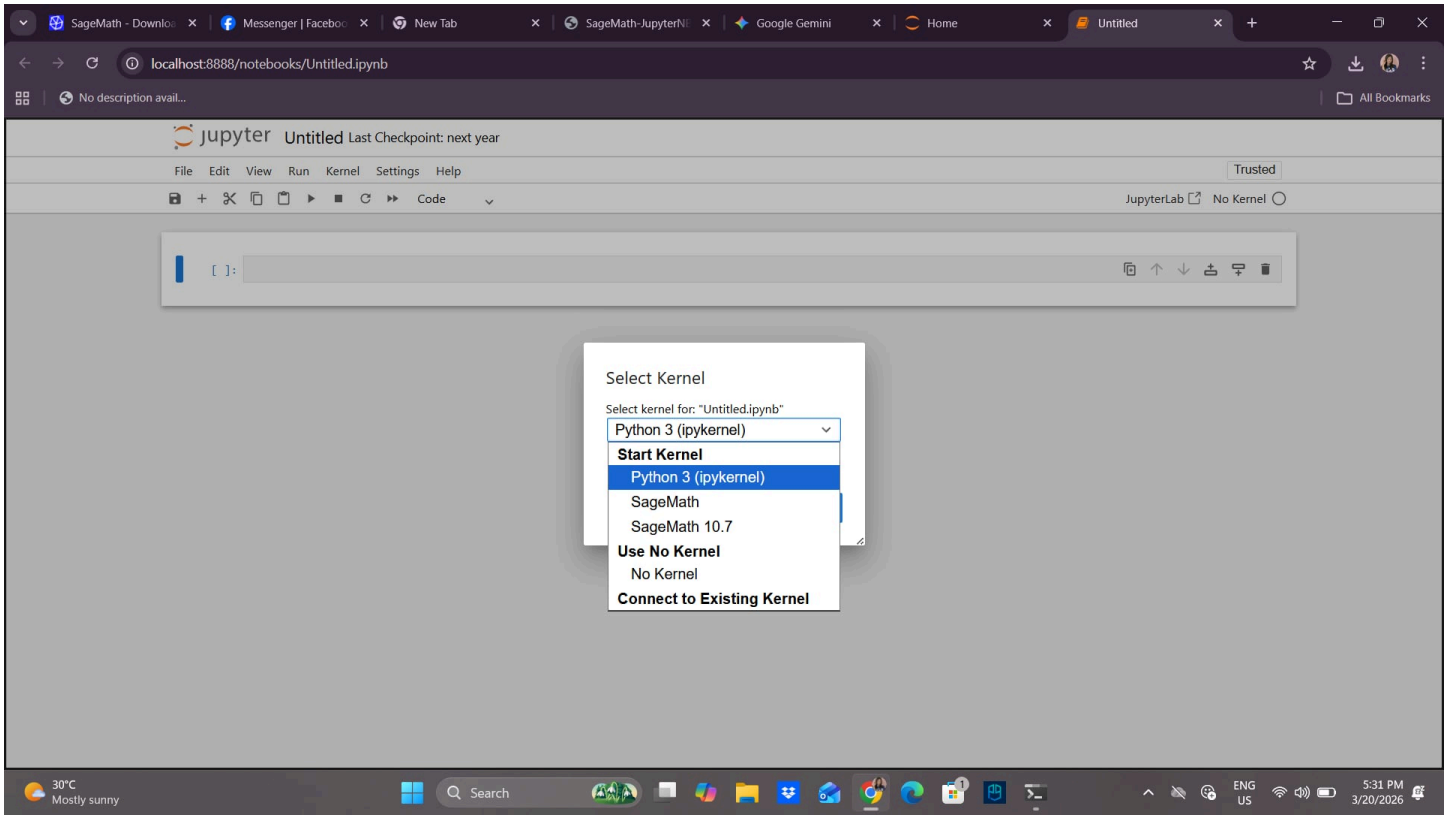
If you're using Jupyter Notebook, this is what you should see:



Click File >> New >> Notebook



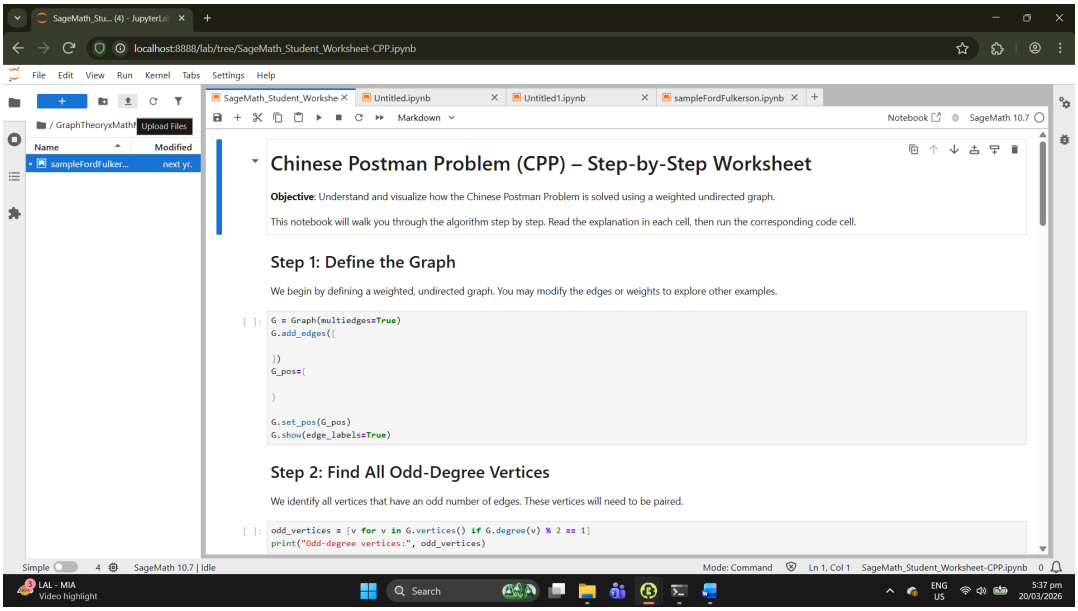
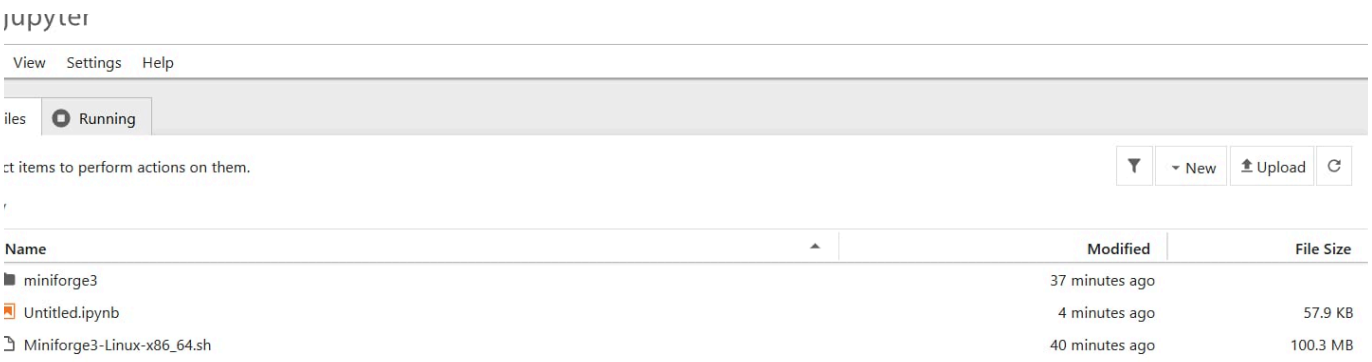
Then you should see this, in which you should select SageMath 10.7



Upload External SageMath File

To upload external files, just click the Upload, then find the file you want to upload:

(If you're in Jupyter Lab)



(If you're in Jupyter Notebook on the upper right)